VSB TECHNICAL UNIVERSITY OF OSTRAVA | FACULTY OF MECHANICAL ENGINEERING | DEPARTMENT OF APPLIED MECHANICS

FABER COST CA23109

cost EUROPEAN COOPERATION IN SCIENCE AND TECHNOLOGY

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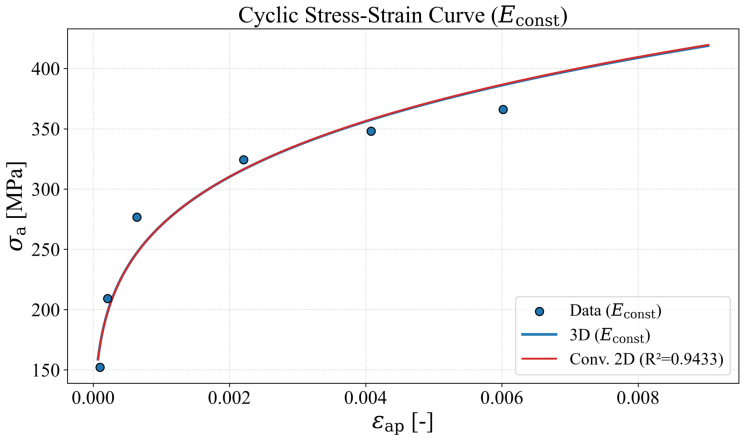
E-N CURVE PROTOCOL - PUSH-PULL (Page 1)

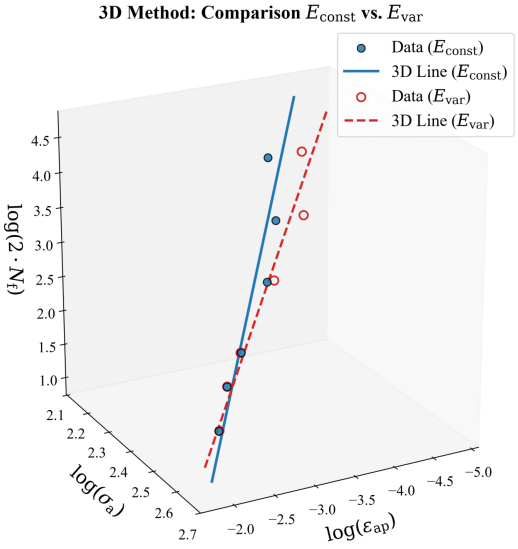
Experimenter:	Radim Halama, VSB-TU Ostrava	**Date:**	2025
Post-processing:	Radim Halama, VSB-TU Ostrava	**Date:**	13.10.2026
Loading type:	Uniaxial AlSi10MgSLMZ, mean strain 0%, Case 0		
Notes:	1%/s, Extensometer - gauge length 10 mm. Criterion: Nf_E10		

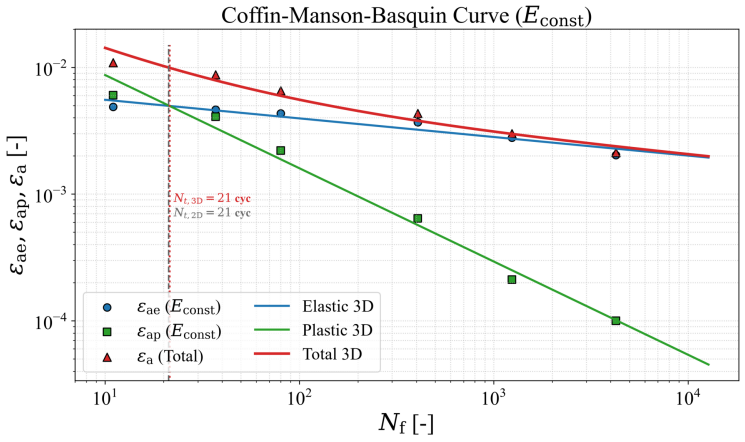
Input Data (Evaluated with $E_{\text{const}} = 75000$ MPa):

ID	ε_a [%]	σ_a [MPa]	N_f [-]	E_{var} [MPa]	w
1	1.09	366.1	11	73404	1
2	0.87	348.0	37	74414	1
3	0.65	324.3	80	75400	1
4	0.43	276.8	406	72671	1
5	0.30	209.2	1240	71926	1
6	0.21	152.2	4253	72700	1

N_f is defined as a 10% decrease in dynamic modulus (N_f^{E10})







Material Parameters (E_{const} / E_{var}):

Parameter	3D (const / var)	Conv. (const / var)
K' [MPa]	1071.6 / 854.9	1077.1 / 859.0
n' [-]	0.1995 / 0.1591	0.2003 / 0.1598
σ'_f [MPa]	645.3 / 651.7	640.4 / 640.4
b [-]	-0.1468 / -0.1484	-0.1455 / -0.1455
ε'_f [-]	0.0787 / 0.1818	0.0769 / 0.1669
c [-]	-0.7357 / -0.9328	-0.7318 / -0.9186

Transition life Nt: 3D = 21 cycles | Conv. = 21 cycles

Reliability R^2 (E_{const} / E_{var}):

R^2	3D (const / var)	Conv (const / var)
CSSC	0.943 / 0.963	0.943 / 0.963
Basquin	0.916 / 0.916	0.916 / 0.916
Cof. − Man.	0.985 / 0.968	0.985 / 0.968

Acknowledgments: Financial support from the Czech Science Foundation / FABER COST Action 23109.

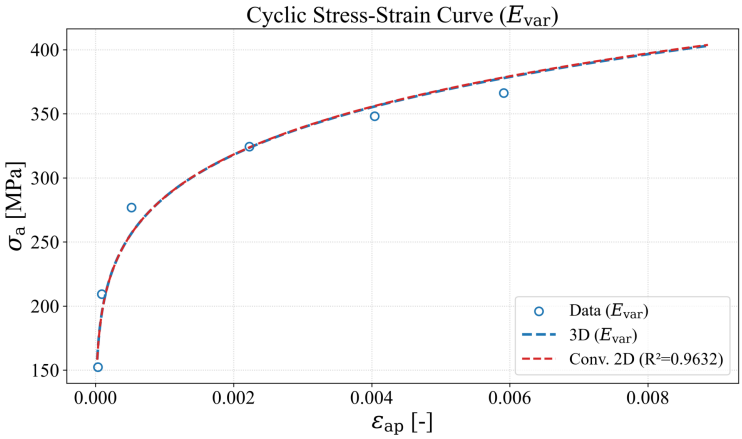
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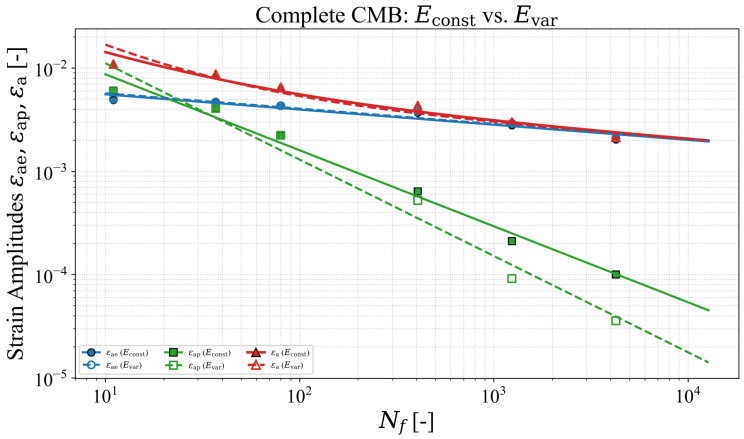
Ref: Radim Halama et al. (2026) "Prediction of ratcheting and LCF damage using a memory surface in cyclic...." International Journal of Fatigue 210: 109658

DOI: https://doi.org/10.1016/j.ijfatigue.2026.109658

E-N CURVE PROTOCOL - ENERGY & COMPARISONS (Page 2)

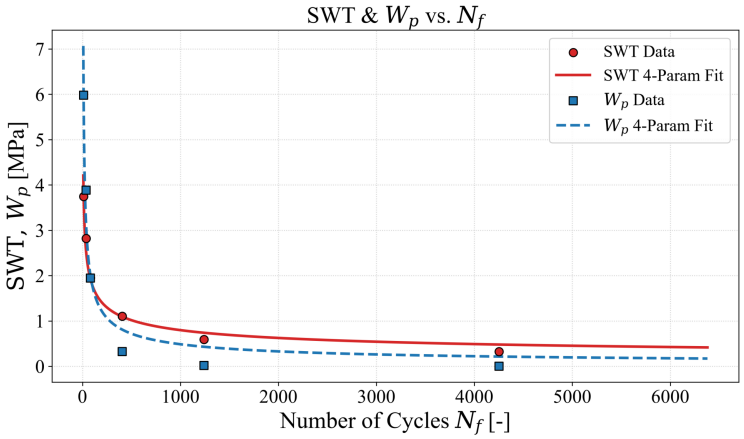
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Specimen Energy & Damage Parameters:

ID	σ_m [MPa]	W_p [MPa]	SWT [MPa]
1	-22.7	5.97689	3.74273
2	-24.7	3.88231	2.81891
3	-25.2	1.94388	1.95286
4	-22.1	0.32624	1.10285
5	-12.8	0.02005	0.58911
6	-0.7	0.00017	0.32291



4-Parametric Fatigue Model ($FP = C_1 N_f^{C_2} + C_3 N_f^{C_4}$):

Fatigue Parameter	C_1	C_2	C_3	C_4	R^2
SWT [MPa]	1.9536e-16	-0.2735	9.0375e+00	-0.3516	0.9836
W_p [MPa]	1.1893e-10	-0.3498	2.4190e+01	-0.5659	0.9643

Ref: Radim Halama et al. (2026) "Prediction of ratcheting and LCF damage using a memory surface in cyclic...." International Journal of Fatigue 210: 109658
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